

CyberShield AI

Month 3 Core Scanners & AI Classifiers

1. Goal

The primary goal of the third month of the **CyberShield AI** project was to develop the core cyber security scanning modules and implement intelligent AI-based classification systems for detecting and analyzing various cyber threats. This phase focused on transforming the project from a basic application into a practical cyber security platform capable of identifying suspicious activities, evaluating security risks, and educating users about modern cyber attacks.

During this month, several essential scanning engines were developed, including the **Website/URL Scanner, Email Phishing Analyzer, Password Strength Analyzer, Malware File Scanner, and Network Scanner**. These modules were designed to analyze different types of cyber threats using industry-standard techniques and security best practices. Each scanner performs detailed security checks, calculates risk scores, and provides recommendations to help users understand potential vulnerabilities.

In addition to the scanning engines, multiple **Artificial Intelligence (AI) classifiers** were implemented to improve the accuracy and efficiency of threat detection. The AI models analyze URLs, phishing emails, scam messages, and various cyber threats using rule-based classification techniques. The system also includes a Threat Explainer that provides easy-to-understand descriptions of detected threats along with their corresponding MITRE ATT&CK techniques and recommended mitigation strategies.

The successful completion of this phase significantly enhanced the functionality of the CyberShield AI platform by introducing intelligent threat analysis, automated risk assessment, and practical cyber security tools. These features provide users with a secure environment to understand cyber attacks, evaluate security risks, and improve their overall cyber security awareness. Furthermore, this development phase establishes the foundation for implementing advanced modules such as attack simulations, security training, and AI-powered assistance in the upcoming stages of the project.

2. Website / URL Scanner

The **Website / URL Scanner** is one of the most important components of the CyberShield AI platform. It is designed to evaluate the safety and legitimacy of websites before users visit them. Cybercriminals frequently use fake or malicious websites to steal sensitive information such as usernames, passwords, banking credentials, and personal data through phishing attacks. Therefore, this module helps users identify suspicious websites and avoid becoming victims of online fraud.

The Website Scanner performs a comprehensive security analysis by executing **thirteen different security checks** on every submitted URL. Each check identifies specific indicators commonly associated with phishing websites, malicious domains, insecure web applications, or deceptive online content. Based on the results of these checks, the system calculates a **risk score** and classifies the website into one of three categories: **Safe, Suspicious, or Dangerous**.

2.1 Website Analysis Workflow

The Website Scanner follows a structured workflow to evaluate each URL:

User enters URL → Security checks are executed → Risk score is calculated → Risk level is assigned → Detailed security report is generated

This automated process enables users to receive immediate feedback regarding the security status of a website before interacting with it.

2.2 Security Checks Performed

The Website Scanner performs the following security checks:

HTTPS Verification

The scanner verifies whether the website uses the HTTPS protocol. Websites that do not use HTTPS transmit data without encryption, making communication vulnerable to interception and Man-in-the-Middle (MITM) attacks.

SSL Certificate Validation

The scanner checks whether the SSL/TLS certificate is valid, trusted, and has not expired. Invalid or self-signed certificates may indicate fake or malicious websites attempting to impersonate legitimate services.

Domain Age Verification

Using WHOIS information, the scanner determines the domain registration date. Newly registered domains are considered more suspicious because phishing websites are often created shortly before attacks begin.

Short URL Detection

The system identifies shortened URLs created using services such as Bit.ly or TinyURL. Since shortened links hide the actual destination, attackers frequently use them to disguise malicious websites.

Typosquatting Detection

The scanner detects domains that closely resemble legitimate websites using character substitutions. For example, replacing the letter "l" with the number "1" in "paypal" to create "paypa1.com" can deceive users into visiting fraudulent websites.

Homograph Attack Detection

The scanner identifies Unicode characters that visually resemble normal English letters. These attacks create deceptive domain names that appear identical to trusted websites while redirecting users to malicious pages.

Suspicious Keyword Analysis

The system searches URLs for suspicious keywords such as **login**, **verify**, **secure**, **update**, **bank**, and **account**, which are commonly used in phishing attacks to create urgency and trick users into providing sensitive information.

IP Address Detection

Instead of using a domain name, some malicious websites directly use an IP address within the URL. This behavior is uncommon for legitimate websites and therefore increases the calculated risk score.

Subdomain Analysis

The scanner examines the number of subdomains contained in the URL. Excessive subdomains may indicate attempts to hide the actual destination or imitate trusted websites.

URL Length Analysis

Very long URLs often contain encoded characters and unnecessary parameters designed to confuse users or bypass security filters. The scanner identifies unusually long URLs as potentially suspicious.

Special Character Analysis

The system checks for excessive usage of characters such as '@', '%', '=', '-', and other symbols commonly found in obfuscated phishing URLs.

Page Content Analysis

The scanner downloads the webpage and examines its HTML content for fake login forms, hidden redirects, password input fields, suspicious JavaScript, and other indicators of phishing activity.

Google Safe Browsing Verification

Finally, the submitted URL is checked against the Google Safe Browsing database to determine whether it has already been reported as malicious, deceptive, or unsafe by Google's security systems.

2.3 Risk Score Calculation

Each security check contributes a predefined number of risk points depending on the severity of the detected issue. The combined score determines the overall safety level of the website.

- **0–34:** Safe
- **35–64:** Suspicious
- **65–100:** Dangerous

This scoring mechanism enables users to understand the overall risk associated with a website quickly and effectively.

2.4 Benefits of the Website Scanner

The Website Scanner provides several important benefits:

- Detects phishing websites before users access them.
- Identifies fake domains and typosquatting attacks.
- Verifies SSL certificates and secure communication.
- Detects suspicious URLs and malicious redirections.
- Calculates an overall website risk score.
- Integrates with Google Safe Browsing for enhanced protection.
- Improves cyber security awareness through detailed security reports.

3. Email Phishing Analyzer

The **Email Phishing Analyzer** is a core module of the CyberShield AI platform designed to detect phishing emails and help users identify fraudulent messages before they become victims of cyber attacks. Phishing is one of the most common cyber threats, where attackers impersonate trusted organizations or individuals to steal sensitive information such as usernames, passwords, banking details, and personal data.

The Email Phishing Analyzer examines both the email header and message content using multiple security checks. It identifies suspicious characteristics, calculates a risk score, and provides recommendations to users for recognizing and avoiding phishing attacks.

3.1 Working of the Email Phishing Analyzer

The Email Phishing Analyzer follows the workflow below:

User uploads email or pastes email content → Email headers are analyzed → Sender authentication is verified → Content is scanned → Risk score is calculated → Security report is generated

This automated process enables users to verify whether an email is legitimate or potentially malicious.

3.2 Security Checks Performed

SPF (Sender Policy Framework) Verification

SPF is an email authentication protocol that verifies whether the sender's mail server is authorized to send emails on behalf of the claimed domain. If SPF validation fails, the email may have been spoofed by an attacker.

DKIM (DomainKeys Identified Mail) Verification

DKIM uses cryptographic signatures to verify that the email has not been modified during transmission. Invalid or missing DKIM signatures increase the probability of phishing or email tampering.

DMARC Policy Verification

DMARC works together with SPF and DKIM to protect domains against email spoofing. The analyzer checks whether the sender's domain has a valid DMARC policy and whether the received email complies with it.

Email Header Analysis

The system examines various email header fields such as:

- From Address
- Reply-To Address
- Return Path
- Received Headers
- Message ID
- Routing Information

Any mismatch between these fields may indicate a phishing attempt.

Content Analysis

The analyzer scans the email body for suspicious elements, including:

- Urgent language
- Threatening messages
- Fake security alerts
- Banking requests
- Password reset requests
- Prize or lottery scams
- Suspicious hyperlinks

These indicators help identify social engineering attacks.

Attachment Analysis

Attachments are checked for dangerous file types such as:

- .exe
- .bat
- .js
- .vbs
- .scr
- Macro-enabled Office documents

Malicious attachments are one of the primary methods used to distribute malware.

Sender Domain Reputation

The analyzer performs a WHOIS lookup to determine the sender domain's registration age. Newly registered domains are often associated with phishing campaigns.

3.3 Risk Assessment

Each suspicious indicator contributes to the overall phishing score.

Risk levels are classified as:

- **Safe**
- **Suspicious**
- **High Risk**
- **Phishing Detected**

The system also explains why the email received a particular risk score, making it educational for users.

3.4 Benefits

The Email Phishing Analyzer helps users:

- Detect phishing emails
- Identify spoofed senders
- Verify email authenticity
- Analyze suspicious attachments
- Improve email security awareness
- Reduce the risk of credential theft

Summary

The Email Phishing Analyzer is an essential security component of CyberShield AI. It combines modern email authentication techniques with intelligent content analysis to identify phishing attacks and educate users about secure email practices.

4. Password Strength Analyzer

The **Password Strength Analyzer** is designed to evaluate the security of user passwords and provide recommendations for creating stronger passwords. Weak passwords remain one of the leading causes of cyber attacks because attackers can easily guess or crack them using automated tools.

This module analyzes password complexity, calculates entropy, estimates password cracking time, detects common password patterns, and checks whether a password has appeared in known data breaches.

4.1 Working Process

The Password Strength Analyzer follows this workflow:

User enters password → Password characteristics are analyzed → Entropy is calculated → Common password database is checked → Crack time is estimated →

Security recommendations are generated

4.2 Password Analysis Techniques

Character Set Analysis

The analyzer checks whether the password contains:

- Lowercase letters
- Uppercase letters
- Numbers

- Special characters

Passwords using multiple character sets receive higher security scores.

Password Length Analysis

Longer passwords are generally more secure because they require significantly more time to crack. Passwords shorter than eight characters are considered weak.

Entropy Calculation

Password entropy measures the randomness of a password.

Higher entropy indicates greater resistance against brute-force attacks.

The analyzer calculates entropy based on password length and available character sets.

Strength Scoring

A score between **0 and 100** is assigned based on:

- Password length
- Character diversity
- Entropy
- Pattern detection
- Dictionary words
- Common passwords

4.3 Common Password Detection

The system compares passwords against a list of commonly used passwords such as:

- 123456
- password
- admin
- qwerty
- welcome

If a match is found, the password receives a very low security score.

4.4 Dictionary Word Detection

Passwords containing complete dictionary words are easier to guess using dictionary attacks.

The analyzer identifies such words and recommends stronger alternatives.

4.5 Sequential Pattern Detection

The analyzer detects predictable patterns such as:

- 123456
- abcdef
- qwerty
- asdfgh

These patterns significantly reduce password security.

4.6 Password Crack Time Estimation

The analyzer estimates how long different attack methods would require to crack the password.

Attack scenarios include:

- Online attacks with rate limiting
- Online attacks without rate limiting
- Offline CPU attacks
- Offline GPU attacks
- High-performance GPU cluster attacks

Users receive realistic estimates of password strength under different attack conditions.

4.7 Have I Been Pwned Integration

The Password Analyzer integrates with the **Have I Been Pwned (HIBP) API** using the **k-Anonymity** privacy model.

Instead of sending the complete password, only the first five characters of its SHA-1 hash are transmitted.

This allows users to determine whether their password has appeared in previous data breaches without revealing the actual password.

4.8 Password Strength Levels

Based on the calculated score, passwords are classified into five categories:

- Very Strong
- Strong
- Moderate
- Weak
- Very Weak

The analyzer also provides suggestions for improving password security.

Benefits

The Password Strength Analyzer helps users:

- Create stronger passwords
- Prevent brute-force attacks
- Detect reused passwords
- Identify breached passwords
- Improve password management practices
- Enhance account security

Summary

The Password Strength Analyzer provides an intelligent method for evaluating password security using entropy calculations, attack simulations, breach detection, and strength scoring. It encourages users to adopt secure password practices and significantly reduces the risk of unauthorized account access.

5. Malware File Scanner

The **Malware File Scanner** is an important security module of the **CyberShield AI** platform. Its primary purpose is to detect malicious files before they are executed on a computer system. Malware such as viruses, worms, trojans, ransomware, spyware, and rootkits can compromise sensitive information, damage systems, or provide attackers with unauthorized access. This module helps users identify potentially harmful files by performing multiple security checks and generating a detailed analysis report.

The scanner uses a combination of **hash-based detection, YARA rule matching, VirusTotal API integration, file type verification, and entropy analysis** to determine whether a file is safe or malicious. This multi-layered approach increases detection accuracy while minimizing false positives.

5.1 Working Process

The Malware File Scanner follows the workflow below:

User uploads a file → File properties are extracted → Cryptographic hashes are generated → File type is verified → YARA rules are applied → VirusTotal lookup is performed → File entropy is calculated → Risk score is generated → Security report is displayed

This process enables users to receive an accurate assessment of the uploaded file without permanently storing it on the server.

5.2 Cryptographic Hash Generation

The scanner generates three cryptographic hashes for every uploaded file:

- **MD5**
- **SHA-1**
- **SHA-256**

These hashes uniquely identify the file and are used to compare it against known malware databases. If the generated hash matches an existing malicious signature, the file is flagged as dangerous.

5.3 YARA Rule Scanning

YARA is a popular malware detection framework used by cybersecurity professionals to identify malicious files based on predefined patterns.

The Malware Scanner compares uploaded files against multiple YARA rules to detect:

- Malware signatures
- Suspicious strings
- Executable patterns
- Packed files
- Known ransomware indicators

If a rule matches, the scanner identifies the type of malware and reports the detected threat.

5.4 VirusTotal Integration

The scanner integrates with the **VirusTotal API** to verify whether the uploaded file has already been analyzed by multiple antivirus vendors.

VirusTotal provides:

- Detection ratio
- Antivirus results
- Community reputation
- Threat categories
- File behavior analysis

Using VirusTotal significantly improves the reliability of malware detection.

5.5 File Extension Verification

Attackers often disguise malicious files by changing their extensions.

For example:

- invoice.pdf.exe
- photo.jpg.exe
- report.doc.scr

The scanner compares the file extension with the actual file format to detect extension spoofing attacks.

5.6 File Entropy Analysis

Entropy measures the randomness of data inside a file.

Very high entropy may indicate:

- Encryption
- Compression
- Packed executables
- Obfuscated malware

The scanner calculates file entropy to detect suspicious characteristics commonly associated with modern malware.

5.7 Secure File Handling

To maintain user privacy and system security:

- Files are scanned entirely in memory.
- Uploaded files are deleted immediately after analysis.
- No permanent file storage is performed.
- Maximum upload size is limited to **16 MB**.

These security measures reduce the risk of accidental malware execution and unauthorized data retention.

5.8 Risk Assessment

The Malware Scanner assigns one of the following classifications:

- Safe
- Suspicious
- Malware Detected
- High Risk

A detailed report explains the reasons behind the assigned classification and provides security recommendations.

Benefits

The Malware File Scanner offers several advantages:

- Detects malicious files before execution.
- Verifies file integrity using cryptographic hashes.
- Integrates with VirusTotal for enhanced threat intelligence.
- Detects disguised executable files.
- Identifies packed and encrypted malware.
- Protects user privacy by deleting uploaded files after analysis.

Summary

The Malware File Scanner provides a comprehensive security analysis of uploaded files using multiple detection techniques. It enables users to identify malware safely, understand potential threats, and reduce the risk of malware infections through early detection and detailed reporting.

6. Network Scanner

The **Network Scanner** is another essential module of the CyberShield AI platform. It is designed to analyze devices connected to a local network, identify open ports, detect running services, and evaluate potential security vulnerabilities. Network scanning plays an important role in cyber security because attackers frequently exploit exposed services and misconfigured network devices.

The Network Scanner provides users with a clear understanding of their network environment by identifying active hosts, open communication ports, and services that may introduce security risks.

6.1 Working Process

The Network Scanner follows the workflow below:

User enters network address or IP range → Devices are discovered → Open ports are scanned → Running services are identified → Risk assessment is performed → Security report is generated

This automated process enables users to quickly identify vulnerable devices within their network.

6.2 Device Discovery

The scanner identifies all active devices connected to the specified network.

For each discovered device, the following information is collected:

- IP Address
- Host Name
- MAC Address
- Operating System (if detectable)
- Device Status

This information helps users understand which devices are connected to their network.

6.3 Port Scanning

The Network Scanner checks commonly used ports to determine whether they are open or closed.

Examples include:

Port	Service
21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
143	IMAP
443	HTTPS
3306	MySQL
3389	Remote Desktop
8080	HTTP Alternative

Each open port is analyzed to determine whether it introduces security risks.

6.4 Service Detection

The scanner identifies services running on open ports.

Examples include:

- Web Servers
- Database Servers
- Mail Servers
- FTP Servers
- SSH Services
- Remote Desktop Services

Knowing which services are exposed helps administrators secure unnecessary or vulnerable services.

6.5 High-Risk Port Detection

Some ports are considered high risk because they are frequently targeted by attackers.

Examples include:

- Telnet (23)
- FTP (21)
- Remote Desktop (3389)
- SMB (445)

If these ports are publicly accessible, the scanner increases the overall risk score and recommends appropriate security measures.

6.6 Risk Assessment

Each device receives an overall security rating based on:

- Number of open ports
- High-risk services
- Unsecured protocols
- Potential vulnerabilities

Devices are classified as:

- Safe
- Moderate Risk
- High Risk
- Critical

6.7 Security Recommendations

Based on the scan results, the Network Scanner recommends actions such as:

- Closing unused ports
- Disabling insecure services
- Enabling firewalls
- Updating software
- Restricting remote access
- Using secure communication protocols

These recommendations help improve overall network security.

Benefits

The Network Scanner enables users to:

- Discover connected devices.
- Identify open communication ports.
- Detect insecure services.
- Improve network security.
- Reduce attack surfaces.
- Monitor local network environments.

Summary

The Network Scanner is a powerful network security assessment tool that helps users identify vulnerable devices, insecure services, and exposed communication ports. By providing detailed security reports and practical recommendations, it enhances network visibility and supports proactive cyber security management.

7. AI Classifiers

Artificial Intelligence (AI) plays a significant role in modern cyber security by enabling automated threat detection and intelligent decision-making. The **CyberShield AI** platform incorporates multiple rule-based AI classifiers that analyze different types of cyber threats and provide accurate security assessments. Instead of relying on complex machine learning models, the platform uses explainable rule-based classifiers that are lightweight, fast, and easy to maintain.

The AI classification system examines URLs, emails, scam messages, and cyber threats by extracting important security features and assigning confidence scores based on predefined rules. These classifiers improve detection accuracy while helping users understand the reasoning behind each security decision.

7.1 URL Classifier

The URL Classifier evaluates website addresses to determine whether they are legitimate or potentially malicious. It extracts multiple features from the submitted URL, including HTTPS usage, domain age, suspicious keywords, URL length, typosquatting, excessive subdomains, IP-based URLs, and SSL certificate validity.

Each feature contributes a weighted score to the final analysis. Based on the total score, the classifier predicts whether the website is **Safe**, **Suspicious**, or **Phishing**. It also provides a confidence percentage and phishing probability to help users understand the level of risk associated with the website.

7.2 Email Classifier

The Email Classifier analyzes email messages to identify phishing attempts and spam. It examines subject lines, sender information, suspicious keywords, embedded links, attachments, and email headers using regular expression (Regex) patterns.

The classifier detects urgency phrases such as *"Verify your account," "Immediate action required," "Congratulations! You have won,"* and similar expressions commonly used in phishing campaigns. It then assigns a confidence score and explains which indicators contributed to the detection.

7.3 Scam Message Classifier

The Scam Message Classifier is designed to detect fraudulent SMS messages, instant messages, and social media scams. It recognizes common scam categories including:

- Prize and lottery scams
- Banking fraud
- Investment scams
- Fake job offers
- Courier delivery scams
- UPI payment scams
- OTP verification fraud

The classifier analyzes suspicious keywords, URLs, phone numbers, payment requests, and urgency indicators. It also identifies scam patterns commonly targeting Indian users, such as fake KYC updates, UPI verification messages, and OTP theft attempts.

7.4 Threat Explainer

The Threat Explainer is an educational AI component that provides detailed explanations of detected cyber threats in simple and easy-to-understand language. Instead of displaying only technical information, the module explains:

- What the threat is
- How the attack works
- Possible consequences
- Recommended preventive measures
- MITRE ATT&CK technique mapping
- Severity level

This feature improves cyber security awareness by helping users understand both the technical and practical aspects of cyber attacks.

7.5 Confidence Score

Each AI classifier generates a confidence score indicating how certain the system is about its prediction. Higher confidence values represent stronger evidence that the detected threat is accurate. This helps users make informed security decisions while improving trust in the analysis results.

Benefits

The AI Classifiers provide several advantages:

- Intelligent threat detection.
- Fast rule-based analysis.
- Explainable decision-making.
- Reduced false positives.
- Educational threat explanations.
- Improved cyber security awareness.
- Easy integration with other security modules.

Summary

The AI Classification System enhances the CyberShield AI platform by providing intelligent, explainable, and automated threat analysis. These classifiers improve the platform's ability to detect phishing attacks, scam messages, malicious URLs, and other cyber threats while educating users about safe online practices.

8. Security Features & Risk Analysis

The CyberShield AI platform implements multiple security mechanisms to ensure accurate threat detection and user safety. Each scanner and AI module contributes to the overall security analysis by identifying suspicious behavior, calculating risk scores, and generating detailed reports.

Risk assessment is based on predefined security rules, threat severity, and confidence levels. Every detected issue increases the overall risk score, allowing users to quickly understand the security status of websites, emails, passwords, files, and network devices.

Risk levels are categorized as:

- **Safe:** No significant security issues detected.
- **Suspicious:** Minor security concerns requiring user attention.
- **Dangerous:** High probability of malicious activity.
- **Critical:** Immediate action recommended due to severe security threats.

The platform also provides mitigation suggestions, enabling users to reduce risks by following recommended security practices. Combining automated scanners with AI-based analysis results in a comprehensive and reliable cyber security assessment.

9. References

The following references were used during the research and development of the core scanning modules and AI classifiers:

9.1 OWASP Top 10

Used to understand common web application vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), Broken Authentication, and Security Misconfiguration.

9.2 MITRE ATT&CK Framework

Studied for mapping cyber attack techniques, threat classification, and defensive strategies.

9.3 Google Safe Browsing API

Used for identifying malicious and phishing websites during URL analysis.

9.4 VirusTotal API

Used for malware detection through hash lookups and multi-engine antivirus analysis.

9.5 Have I Been Pwned (HIBP) API

Integrated into the Password Strength Analyzer to identify passwords exposed in previous data breaches using the k-Anonymity privacy model.

9.6 WHOIS Database

Used to obtain domain registration information and determine the age of websites during phishing detection.

9.7 YARA Documentation

Studied for creating malware detection rules and identifying malicious file patterns.

9.8 Flask Official Documentation

Used as the primary development reference for implementing Flask applications, Blueprints, routing, authentication, and REST APIs.

Summary

These references provide industry-standard guidance and technical knowledge, ensuring that the CyberShield AI platform follows recognized cyber security best practices.

10. Conclusion

The third month of the **CyberShield AI** project successfully completed the development of the core cyber security scanning engines and intelligent AI classification modules. Major functionalities including the **Website/URL Scanner**, **Email Phishing Analyzer**,

Password Strength Analyzer, **Malware File Scanner**, **Network Scanner**, and **AI-based threat classifiers** were implemented and integrated into the platform.

Each module performs specialized security analysis, calculates risk scores, and generates detailed reports to help users identify cyber threats effectively. The implementation of AI-based classification further improves detection accuracy by providing confidence scores, threat explanations, and practical security recommendations.

The successful completion of these modules transforms the CyberShield AI platform into a functional cyber security awareness and threat detection system. Users can now analyze websites, emails, passwords, files, and network devices while learning about common cyber attacks through detailed explanations and intelligent recommendations.

In Month 4,
the project will focus on developing advanced features, including the **Attack Simulator**, **Security Training System**, **Threat Intelligence Module**, and **AI Security Assistant (Chatbot)**. These additions will further enhance the educational value and practical capabilities of the CyberShield AI platform, making it a comprehensive cyber security learning and analysis solution.